

Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset

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Research Goal

Enable state-of-the-art LLM pretraining over long token horizons (15T+) by developing methods that preserve data quantity without sacrificing quality. Current aggressive filtering removes 90% of data, creating a bottleneck for extended training runs.

NEMOTRON-CC ABSTRACT

Metrics: MMLU = Massive Multitask Language Understanding; Score = percentage of correctly answered multiple-choice questions (0-100 scale) - **Problem:** Aggressive filtering in FineWeb-Edu, DCLM removes ~90% of data → unsuitable for long-horizon training (15T tokens) - **Solution:** Classifier ensembling + synthetic rephrasing + reduced heuristic filtering - **Results:** - Short horizon (1T tokens, 8B): - High-quality subset: +5.6 pp MMLU vs. DCLM - Long horizon (15T tokens, 8B, 7.2T from Nemotron-CC): - vs. Llama 3.1 8B: +5.0 pp MMLU (70.3 vs. 65.3) - +3.1 pp ARC-Challenge - +0.5 pp average (10 tasks) - Data efficiency (1T tokens): - Full dataset: parity MMLU with DCLM, but 4× more unique real tokens

- **Composition (6.3T total):**

- Deduplicated original tokens: 4.4T
- Synthetic tokens: 1.9T

INTRODUCTION

- Context: Internet crawl = largest source of unique tokens for LLMs
 - Two purposes: high-quality content + diversity
- Status quo (FineWeb-Edu, DCLM):
 - Use model-based filters for high-quality content
 - Boost benchmark accuracies
 - Cost: remove ~90% of data
- Problem - Long-horizon training bottleneck:
 - Recent models train on extended token horizons
 - Llama 3.1 (8B–405B): 15T tokens
 - Gemma 2 27B: 13T tokens
 - Aggressive pruning creates data scarcity for extended training
 - DCLM & FineWeb-Edu contain ~80% near-duplicates (1T and 0.2T unique tokens respectively)
 - Training many epochs on same data = diminishing returns (Muennighoff et al., 2024); better to train on more unique tokens
- Proposed solution: Better accuracy-quantity trade-off via:
 - Classifier ensembling
 - Synthetic data generation
 - Reduced heuristic filter reliance
- Contributions:
 - 6.3T token long-horizon dataset: 4.4T deduplicated original + 1.9T synthetic
 - Released under Common Crawl Terms of Use (CC contributor framework)
 - NeMo Curator library (Apache 2.0): open-source data curation toolkit with reference implementation
 - Quality classifier models released
 - Effectiveness proof vs. DCLM/FineWeb-Edu:
 - 1.1T high-quality subset: +5.6 pp MMLU
 - Full dataset: parity MMLU vs. DCLM, but 4× more unique real tokens
 - 8B @ 15T tokens: +5.0 pp MMLU vs. Llama 3.1 8B (70.3 vs. 65.3)
 - Ablations show:
 - Ensembling selects larger, more diverse high-quality tokens
 - Rephrasing reduces noise + produces fresh variants

- Disabling heuristic filters boosts high-quality token yield
- Guiding principle: Shift from static heuristic pipeline → learned flywheel
 - Improved data → better LLMs → better synthetic data & quality classification (virtuous cycle)

METHODS

- Overview: Three-component approach to build English Common Crawl pretraining dataset
 - HTML-to-text extraction & heuristic filters (token yield)
 - Model-based quality labeling pipeline
 - Synthetic data generation
- Full pipeline schematic: Figure 3 (Appendix A)

2.1 HTML-to-text Extractor & Filter

- Foundation: Text extraction from HTML is major source of pretraining data
- Reference standard: FineWeb-Edu classifier (model-based quality classifier, proven effective)
- Goal: Balance data quality against token yield, avoid removing good tokens via over-filtering

HTML-to-text Extraction

- Tested: Justext (Pomikálek, 2011) vs. Trafilatura (Barbaresi, 2021)
- Qualitative assessment: both same level
- Quantitative (13 Common Crawl snapshots, Table 1):
 - Justext yields more tokens overall
 - Justext high-quality tokens: +28.6% (by FineWeb-Edu standard: scores 3–5)
 - Strategy: Maximize absolute HQ token count, not percentage
 - Justext: 127B HQ tokens (7.0%)
 - Trafilatura: 104B HQ tokens (7.5%)
- Language filtering: pylld2 + FastText language identifier (threshold 0.3)
- Deduplication:
 - Global fuzzy deduplication
 - Exact substring deduplication across eighths of snapshots
 - Tools: NeMo Curator library + deduplicate-text-datasets library

Filtering

- Conventional heuristic pipeline (from Parmar et al., 2024):
 - Raffel et al. (2020) + Rae et al. (2021) heuristics
 - KenLM perplexity filter (trained on Wikipedia + books)
- Problem: Heuristic filters remove 18.1% of high-quality tokens (FineWeb-Edu standard)
- Proposal: Apply heuristic filters selectively
 - Do NOT apply to high-quality tokens (identified by model-based classifiers)
 - Apply only to low-quality splits
- Validation: Ablation studies (Section 3.3)

Model-based Quality Labeling

- Problem: Existing classifiers (DCLM, FineWeb-Edu) have limited recall (~10% high-quality tokens)
 - Bottleneck for long-horizon training
 - Quality labels not necessarily aligned with LLM downstream task performance
- Solution: Ensemble-based quality labeling pipeline
 - Build three classifiers with different quality preferences
 - Ensemble scores across all documents
 - Split corpus into quality buckets
 - Regroup into 5 quality levels based on downstream task performance

Quality Classifier Training

- Two annotation versions: prompt Mistral 8x22B-instruct + Nemotron-340B-instruct
 - Score documents from FineWeb on educational value (0–5 scale)
- Fine-tune linear regression models on Snowflake-arctic-embed-m embeddings (text-to-vector conversion model)
 - Frozen embedding + encoder layers (weights unchanged during training, only linear layer updated)
 - 20 epochs, learning rate 0.0003
 - Checkpoint selected by highest F1 on validation set
- Third classifier: DCLM classifier (fastText-based, Li et al., 2024)
 - Trained on instruction-formatted data + ELI5 subreddit high-scoring posts (Explain-Like-I'm-5: community-curated, high-quality explanations)

- Stronger performance identifying high-quality tokens vs. FineWeb-Edu
- Offers different perspective for increased recall

Quality Scoring and Bucketing

- Each classifier predicts quality scores for all documents
- Round scores to integers 0–19 (20 equal-sized buckets, ~5% of documents per bucket)
 - Bucket 19 = top 5% highest quality
- Final quality score: ensemble via maximum operation across three classifiers
 - Document distribution across buckets skewed by ensemble operation

Quality Labeling

- Map fine-grained scores (0–19) to 5 downstream quality categories
- Method: continuous pretraining annealing
 - Setup: 50B token training on 70% pretrained 8B model
 - Metric: average performance across 9 tasks per bucket
 - Weighted mix: 66% standard baseline data + 34% evaluated bucket
 - (Tests each bucket's contribution; higher gain = higher quality category)
- Group 20 buckets into 5 quality categories based on performance

Synthetic Data Generation

- Observation: Low-quality docs contain noise/errors; high-quality have good writing/formatting
- Strategy: Different approaches for low vs. high quality

Low-Quality Data

- Goal: Reduce noise/errors while preserving useful information (cut training compute)
- Method: Rephrasing using medium-sized LM (Maini et al., 2024)
 - NOT creating new knowledge (unlike textbook/story generation approaches)
 - Transform text to Wikipedia style (Prompt 5, Appendix H)
 - Reduces errors, redundancies, improves formatting
- Result: Enhanced parallel corpus, lower perplexity, better downstream performance

High-Quality Data

- Goal: Generate unique tokens, condense essential knowledge
 - Rationale: Repeated tokens show diminishing returns after 4 epochs (Muennighoff et al., 2024)
- Four synthesis prompts:
 - Diverse Question-Answer pairs: ask varied question types (yes/no, open-ended, multiple-choice) with correct answers
 - Distill: rewrite as concise, clear passage
 - Extract knowledge: rewrite only informative content, discard filler
 - Knowledge list: structure key information as organized list
- Requirements: Clear, concise responses preserving factual info + concrete details (numbers, etc.)

Processing

- Chunking: Split documents into segments (complete lines, token limit enforced)
 - Prevents over-simplified model outputs on long inputs
 - Discard over-length lines exceeding token limit
- Post-processing:
 - Remove incomplete results, markdown formatting (e.g., **bold**)
 - Strip prefixes (“Here is a paraphrased version:”, “Paraphrased Text:”)
 - Remove enclosing quotation marks
 - Filter under-length outputs (< 50 tokens)
 - Wikipedia results: concatenate passages from same original document
 - Diverse QA results: shuffle pairs, append to segment end

Scale & Implementation

- Model: Mistral NeMo 12B (instruct, FP8 inference)
 - top-p: 0.9, temperature: 0.5
- Synthesis (Table 3):
 - Low-quality (403.0B raw) → Wikipedia → 336.3B synthetic
 - High-quality (451.3B raw):
 - Wikipedia: 372.9B
 - Diverse QA Pairs: 499.5B
 - Distill: 157.6B

- Extract Knowledge: 303.6B
- Knowledge List: 203.2B
- Total synthetic: 1.8T tokens
- Medium-quality: not synthesized (resource constraints)
- Tools: TensorRT-LLM + NeMo-Skills

Putting It All Together

- Combine all techniques above on 99 Common Crawl snapshots (CC-MAIN-2013-20 through CC-MAIN-2024-30)
- **Nemotron-CC dataset: 6.3T tokens**
 - 4.4T globally deduplicated original tokens
 - 1.9T synthetically derived tokens
- Comparison to baselines:
 - ~4× more unique tokens vs. FineWeb-Edu-2 and DCLM
 - Reason: DCLM/FineWeb used sharded approximate deduplication only, contain ~80% fuzzy duplicates
- High-quality subset (for fairer short-horizon comparison):
 - Nemotron-CC-HQ: 1.1T tokens
 - Composition: highest-scoring real data + diverse QA pair synthetic variants only

Dataset sizes in trillions of tokens. “Unique” shows the estimated number of tokens after global fuzzy deduplication of the real tokens.

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EXPERIMENTS

Experiment Setup

Training Setup

- Framework: Megatron-LM library (open source)

- Model: 8B parameter transformer LLMs
- Hyperparameter details: Appendix D

Data Blend

- Default: 1T tokens training
- Composition:
 - 73% English Common Crawl data (variable across experiments)
 - 27% fixed specialized mix: code, papers, books, patents, Wikipedia (Adler et al., 2024)
- Comparison method: vary only CC portion, keep specialized mix constant

Evaluation Setup

- Framework: LM Evaluation Harness library (open source)
- Tasks (10 total, metrics in parentheses):
 - ARC-Easy / ARC-Challenge (normalized accuracy): Multiple-choice science questions; Easy vs. challenging difficulty levels
 - Hellaswag (normalized accuracy): Predict next sentence in video paragraph descriptions
 - Winogrande (accuracy): Pronoun resolution under ambiguous context
 - RACE (accuracy): Reading comprehension from exam passages
 - PIQA (normalized accuracy): Physical intuition / common-sense reasoning about physical interactions
 - Social IQA (accuracy): Understanding social situations and appropriate responses
 - Commonsense QA (accuracy): Multiple-choice commonsense reasoning questions
 - Openbook QA (normalized accuracy): Fact-based Q&A with reference text provided
 - MMLU (accuracy): Broad multitask language understanding (57 diverse subjects: science, humanities, social science, etc.)

Main Results

Short Token Horizon (1T tokens)

- Setup: 8B parameter models trained for 1T tokens
- Results: Table 5

Table 5: Results for 8B parameter models trained on 1T tokens (73% English Common Crawl from the tested dataset, 27% the same, fixed non-Crawl datasets). The models were evaluated on ARC-Easy, ARC-Challenge, Hellaswag,

Winogrande, RACE, PIQA, Social IQA, Commonsense QA, Openbook QA, and MMLU.

Table 5: Results for 8B parameter models trained on 1T tokens (73% English Common Crawl from the tested dataset, 27% the same, fixed non-Crawl datasets). The models were evaluated on ARC-Easy, ARC-Challenge, Hellaswag, Winogrande, RACE, PIQA, Social IQA, Commonsense QA, Openbook QA, and MMLU.

- Nemotron-CC-HQ (high-quality subset):
 - MMLU: +5.6 pp vs. DCLM
 - Average (10 tasks): +3.1 pp vs. DCLM
 - Outperforms on all tasks except RACE
 - Shows effectiveness: classifier ensembling + synthetic data works even without data constraints
- Nemotron-CC (full 6.3T):
 - MMLU: parity with DCLM
 - Average: parity with DCLM
 - Advantage: 4× more unique real tokens
 - Expectation: superior performance in data-constrained settings (long-horizon training)

Long Token Horizon (15T tokens)

- Setup: 8B parameter model trained for 15T tokens
 - 7.2T tokens from Nemotron-CC dataset
- Results: Table 6

Table 6: Comparison of our 8B parameter model vs Llama 3.1 8B. Both were trained for 15T tokens. The numbers for Llama 3.1 are from our own lm-evaluation-harness setup described in Section 3.1 and may not match Meta’s publicly reported numbers, as Meta made various customizations to the benchmarks.

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- vs. Llama 3.1 8B (also trained 15T tokens):
 - MMLU: 70.3 vs. 65.3 (+5.0 pp)
 - Average accuracy: higher overall
 - Conclusion: Dataset proven suitable for state-of-the-art long-horizon training
- Details: Appendix E

3.3 Ablation Study

Extractor & Filter Comparison

- Setup: $4 \times 8\text{B-1T}$ experiments, vary HTML-to-text extractor + heuristic filter application
- Results: Table 7

Table 7: Ablation studies on extractor and filter. HQ means high-quality data judged by FineWeb-Edu classifier (score 3,4,5). HQ-unfiltered means filtering is applied only to LQ data. See Appendix G for more details.

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- Extractors (HTML-to-text conversion):
 - Justext vs. Trafilatura: Justext yields +57.4% more high-quality tokens overall
 - Both qualitatively similar, but Justext quantitatively superior
- Heuristic Filters (automatic rules: length checks, symbol-to-word ratio, KenLM perplexity):
 - Standard application removes 18.1% of HQ tokens (excessive)
 - Proposal: selective application (remove from HQ only, keep on LQ/Medium)
- Downstream impact:
 - Trafilatura filtered vs. Justext unfiltered: marginal differences
 - Justext unfiltered vs. Justext HQ-unfiltered (filters removed from HQ only): +2.0 pp MMLU
 - Conclusion: Preserving HQ tokens by skipping filters improves accuracy

- Decision: Remove heuristic filters from HQ tokens only (maximize HQ token yield without accuracy loss)

Classifiers Comparison

- Three-classifier ensemble:
 1. Custom-trained (Mistral 8x22B annotations, linear regression on Snowflake-arctic-embed-m)
 2. Custom-trained (Nemotron-340B annotations, linear regression on Snowflake-arctic-embed-m)
 3. DCLM classifier (fastText-based, trained on instruction data + ELI5 subreddit)
- Question: Does ensembling improve HQ document recall?
- Classifier overlap analysis (CC-MAIN-2021-21, Table 8):
 - All three predict HQ: 10%
 - FineWeb-Edu (custom variant 1) only: 35.4%
 - DCLM only: 54.4%
 - Insight: Different classifiers capture complementary quality signals
- Downstream performance (13 snapshots, Table 9):
 - HQ token percentage: 9% (single classifier) → 25% (ensemble of three classifiers)
 - Average task performance: on-par or slightly better vs. individual classifiers
 - Conclusion: Ensemble increases HQ token recall while maintaining language understanding performance

Evaluating Synthetic Data

- Setup: $4 \times 8\text{B-1T}$ models, different data blends (Table 10)
 - LQ-Base: original low-quality docs only (baseline)
 - LQ-Synthetic: LQ docs rephrased to Wikipedia style (noise/error reduction, same token count)
 - HQ-Base: 8 epochs high-quality + lower proportion LQ/Medium (quality-heavy blend)
 - HQ-Synthetic: HQ-Base with 4/8 HQ epochs replaced by synthetic variants (QA, Distill, Extract Knowledge, Knowledge List)
- Low-quality rephrasing (LQ-Base vs. LQ-Synthetic):
 - Average gain: +1.50 pp
 - Strong gains: +1.80–4.75 pp on ARC-Easy, ARC-Challenge, OpenBook QA, CommonsenseQA
 - Minor drops on some tasks (potential synthesis artifacts); flagged for future refinement

- High-quality synthetic swap (HQ-Base vs. HQ-Synthetic):
 - Replacing 4/8 repeated HQ epochs with synthetic variants improves most benchmarks
 - Hypothesis: Fresh unique tokens + diverse styles improve learning of specific abilities (e.g., QA, knowledge absorption)

RELATED WORK

- Phi series: Small high-quality data + synthetic, but short token horizons
- FineWeb-Edu & DCLM: Core comparison points; model-based filtering established
 - Nemotron-CC builds on this, improves filtering + data quantity
- Other CC datasets (C4, DOLMA, Gopher, RefinedWeb, TxT360): Extraction + non-learned heuristics
 - RefinedWeb (Penedo et al., 2023): 600B token web-only dataset
 - First to challenge multi-source curation paradigm
 - Aggressive fuzzy + exact deduplication (77% data removal)
 - Demonstrated web-only models competitive with curated corpora (Falcon models)
 - Limitation: High removal rates create bottleneck for long-horizon training
 - Nemotron-CC extends this: Reduces aggressive filtering, adds learned classifiers + synthetic augmentation
 - Other CC datasets: Similar extraction/heuristic emphasis
 - Nemotron-CC: Goes beyond with learned techniques instead
- Concurrent work (Zyda-2): Combines existing datasets (FineWeb-Edu, DCLM, Dolma-CC)
 - Nemotron-CC: Creates new dataset from scratch rather than combining
- Multilingual datasets: Out of scope; future work
- Synthetic data for pretraining:
 - Typical: Instruction pairs, stories, textbooks via GPT-3.5/4 (expensive LMs)
 - Key finding (Maini et al., 2024): Smaller models (Qwen-1.8B, Mistral-7B) adequate for rephrasing
 - Nemotron-CC: Adopts compact-model rephrasing, extends with specialized prompts for LQ vs. HQ

CONCLUSION

- Main achievement: Improved accuracy-quantity trade-offs for long-horizon English CC pretraining
 - Key techniques:
 - Ensembling model-based quality filters
 - Rephrasing low and high-quality documents
 - Reducing reliance on non-learned heuristics
 - Public dataset: Split by quality level + type (real vs. synthetic variants)
 - Enables future work: Quality vs. diversity experiments, short/long-horizon curricula design
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LIMITATIONS

- Model-based ensembling & bucketing:
 - Single strategy explored (resource constraints)
 - Possible improvements at high-quality end of spectrum
- Synthetic rephrasing:
 - No verification of factual accuracy/fidelity to originals
 - Risk of hallucinations or diversity loss (undocumented)
 - Only LQ + HQ rephrasing; Medium-quality excluded
 - Mitigation strategies needed for future work
- Pipeline ablations incomplete:
 - Language identification not thoroughly ablated
 - Likely room for improvement
- Language scope:
 - English only; multilingual extension deferred
- Decontamination:
 - Not performed (no consensus on best practice, impact uncertain)
 - Baselines (FineWeb-Edu, DCLM, Llama 3.1) also undecontaminated
 - DCLM analysis suggests contamination minor factor (MMLU increases post-decontamination)
 - Open question: Impact across different model sizes and token horizons (community research needed)

| Open questions

Experiments -> Main Results -> Short Token Horizon

- Dataset selection mechanism unclear: How were 1T tokens selected from 6.3T pool?